Generics

Generics

- Java Generics allows us to create a single class, interface, and method that can be used with different types of data.
- can create a class that can be used with any type of data. Such a class is known as Generics Class.
- can also create a method that can be used with any type of data. Such a class is known as Generics Method.
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Generics

- **genericsMethod.**
- Public <T> void genericMethod(T data) {...}
- Type parameter <T> is inserted after the modifier public and before the return type void.

Generics method

```
Class Main {
  public static void main(String[] args) {
    // initialize the class with Integer data
    DemoClass demo = new DemoClass();
    // generics method working with String
    demo.<String>genericsMethod("Java Programming");
    // generics method working with integer
    demo.<Integer>genericsMethod(25);   }}
class DemoClass {
  // creae a generics method
  public <T> void genericsMethod(T data) {
    System.out.println("Generics Method:");
    System.out.println("Data Passed: " + data);}}
```

Lambda expression

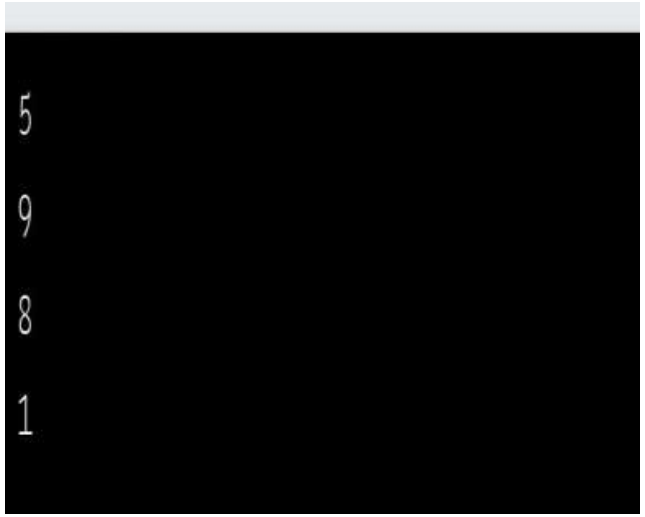
- Lambda expression is a short block of code which takes in parameters and returns a value.
- Lambda expressions are similar to methods, but they do not need a name and they can be implemented right in the body of a method.

Lambda expression

- Simplest lambda expression contains a single parameter and an expression:
- parameter -> expression
- . If the lambda expression needs to return a value, then the code block should have a `return` statement.
- *(parameter1, parameter2) -> { code block}*

Lambda expression

```
Import java.util.ArrayList;
public class Main {
  public static void main(String[] args) {
    ArrayList<Integer> numbers = new
ArrayList<Integer>();
    numbers.add(5);
    numbers.add(9);
    numbers.add(8);
    numbers.add(1);
    numbers.forEach( (n) -> { System.out.println(n); }
    );
  }
}
```



Expression lambda.

- A lambda expression with an expression on the right side is called an expression lambda.
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Block lambdas

- Lambda may need more than a single expression in its lambda body. Java supports Expression Lambdas which contains blocks of code with more than one statement. We call this type of Lambda Body a “Block Body“. Lambdas that contain block bodies can be known as “Block Lambdas“.

Method reference

- `<Class-Name>::<instanceMethodName>`
- Method reference is a shorthand syntax for writing lambda expressions that refer to an existing method.
- Instead of defining a lambda expression that calls a method, you can use method reference to refer to the method directly.

Constructor reference

Also refer constructor

- <Class-Name>::new